Thermal physics — revision exercises

3A — from temperature to mixtures — November 2026

Data used in this sheet (unless the text gives other values).
Specific heat capacity: water $c = 4200\,\text{J/(kg}\,^\circ\text{C)}$ ice $c = 2100\,\text{J/(kg}\,^\circ\text{C)}$ aluminium $c = 900\,\text{J/(kg}\,^\circ\text{C)}$
Specific latent heat: melting $L_f = 3.3 \times 10^5\,\text{J/kg}$ boiling $L_v = 2.3 \times 10^6\,\text{J/kg}$
Linear expansion coefficient: steel $\alpha = 12 \times 10^{-6}/^\circ\text{C}$ aluminium $\alpha = 23 \times 10^{-6}/^\circ\text{C}$ glass $\alpha = 9 \times 10^{-6}/^\circ\text{C}$
Volume expansion: $\gamma \approx 3\alpha$ for solids mercury $\gamma = 1.8 \times 10^{-4}/^\circ\text{C}$
$g = 9.8\,\text{N/kg}$ $T/\text{K} = \theta/^\circ\text{C} + 273$
Newton's law of cooling: $T - T_{\text{room}} = (T_0 - T_{\text{room}})\,e^{-kt}$, so $\ln(T - T_{\text{room}}) = \ln(T_0 - T_{\text{room}}) - k\,t$.

How to use it. Three levels, 25 exercises. *Level 1*: one idea per question, to check that the words and the basic formulas are solid. *Level 2*: standard exam questions of two or three steps. *Level 3*: several ideas combined, like the last question of a Paper 4 or a Paper 6 question with data. Graphs are always drawn by hand, with a ruler and a sharp pencil. Unless the text says otherwise, no energy is lost to the container or to the surroundings.

Level 1 — Foundations

Exercise 1 — Kelvin and Celsius

(a) Convert 37 °C to kelvin, and 250 K to degrees Celsius.

(b) A liquid is heated from 20 °C to 50 °C. Give the temperature change in °C and in K.

(c) A student says: "40 °C is twice as hot as 20 °C". Use kelvin to show that the statement is wrong.

(d) Why can a temperature in kelvin never be negative?

Exercise 2 — Thermometers

1. Thermometers P and Q are identical, except that the capillary tube of P is narrower. Which statement is correct?

   A They have the same sensitivity, because they contain the same liquid.
   B P is more sensitive: for the same expansion the liquid is pushed further along a narrower tube.
   C Q is more sensitive, because a wider tube holds more liquid.
   D P has the larger range.

2. A thermometer placed in pure melting ice reads 1.0 °C. It then reads 38.5 °C in a patient. What is the correct temperature of the patient?

   A 37.5 °C
   B 39.5 °C
   C 38.5 °C
   D 38.0 °C

3. Why is a mercury-in-glass thermometer unsuitable for measuring −50 °C? (Mercury freezes at −39 °C and boils at 357 °C.)

A Mercury boils at $-50\,^{\circ}\mathrm{C}$.

B Mercury would already be frozen at $-50\,^{\circ}\mathrm{C}$.

C Mercury contracts too little.

D The glass would crack.

4. A thermometer scale runs from $-10\,^{\circ}\mathrm{C}$ to $110\,^{\circ}\mathrm{C}$, with a division every $1\,^{\circ}\mathrm{C}$. What are its range and its smallest scale division?

A range $1\,^{\circ}\mathrm{C}$, smallest division $120\,^{\circ}\mathrm{C}$

B range $100\,^{\circ}\mathrm{C}$, smallest division $1\,^{\circ}\mathrm{C}$

C range $120\,^{\circ}\mathrm{C}$, smallest division $1\,^{\circ}\mathrm{C}$

D range $120\,^{\circ}\mathrm{C}$, smallest division $10\,^{\circ}\mathrm{C}$

Exercise 3 — A steel rail

A steel rail is 12.0 m long at $10\,^{\circ}\mathrm{C}$. In the sun it reaches $40\,^{\circ}\mathrm{C}$.

(a) Calculate the increase in length.

(b) What is the new length of the rail?

(c) By how much would a 24.0 m rail expand under the same change of temperature?

(d) Why are gaps left between rails?

Exercise 4 — The particle model

1. Why does a liquid cool when it evaporates?

A The particles become smaller.

B Heat is created by the surroundings.

C The fastest particles escape, so the average kinetic energy of the rest falls.

D The liquid becomes denser.

2. Smoke particles, seen under a microscope, move along zigzag paths. Why?

A They are hit unevenly by fast, randomly moving air molecules.

B They repel each other.

C They are alive.

D They are pushed only by convection currents.

3. A gas is heated in a rigid, sealed container. Why does its pressure rise?

A The particles become larger.

B The particles weigh more.

C The particles repel the walls.

D The particles move faster, so they hit the walls harder and more often.

4. A gas in a syringe is compressed to half its volume, slowly, at constant temperature. What happens to the particles?

A Their average speed doubles.

B Their average speed is unchanged, but they hit the walls about twice as often.

C They become smaller.

D They stop moving.

Exercise 5 — Energy to warm water

(a) How much energy is needed to warm 2.0 kg of water from 15 °C to 35 °C?

(b) A 2.0 kW heater supplies this energy. How long does it take?

(c) The same energy is given to 2.0 kg of cooking oil, with $c = 2000\,\mathrm{J/(kg\,°C)}$. By how many degrees does the oil warm?

(d) Explain why the oil warms more than the water.

Exercise 6 — Energy transfer by heat

1. The heating element of a kettle is at the bottom. Why?

   A Hot water is denser than cold water, so it sinks.
   B Water conducts energy very well, so the position does not matter.
   C Hot water rises and cooler water sinks: convection currents spread the energy.
   D Radiation from the element heats the water at the top.

2. Which feature of a vacuum flask reduces the transfer of energy by *radiation*?

   A The vacuum between the walls.
   B The silvered surfaces of the walls.
   C The plastic stopper.
   D The glass walls.

3. A metal spoon in hot soup becomes hot at the handle, a wooden spoon does not. Why?

   A Metals are good conductors: free electrons carry energy along the spoon.
   B The metal was warmer at the beginning.
   C Metals are better radiators.
   D Convection occurs inside metals.

4. Double glazing keeps a room warm because it traps a layer of air between two panes. Why does this reduce the energy loss?

   A Air is a good conductor.
   B The air between the panes is a vacuum.
   C Air radiates infrared strongly.
   D Air is a poor conductor and, trapped, cannot circulate.

Exercise 7 — Melting and boiling

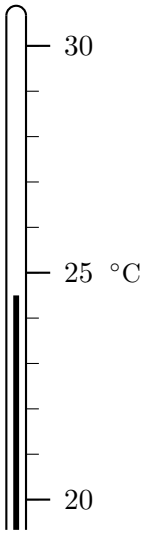
(a) How much energy is needed to melt 0.50 kg of ice at 0 °C?

(b) How much energy is needed to boil away 0.20 kg of water at 100 °C?

(c) Compare the energy needed to boil 1 kg of water with that needed to melt 1 kg of ice.

(d) While ice melts, energy is supplied but its temperature stays at 0 °C. Where does the energy go?

Exercise 8 — Reading a thermometer



(a) Read the thermometer, to the nearest half division.

(b) When placed in melting ice the same thermometer reads 1.0 °C. What is the zero error, and what is the corrected reading in (a)?

(c) Four readings of the same temperature give 24.5 °C, 24.0 °C, 26.0 °C, 24.5 °C. Which one is anomalous? Calculate the mean of the others and correct it for the zero error.

Exercise 9 — Is it a straight line?

A cup of tea cools in a room at $T_{\text{room}} = 20.0\,^{\circ}\text{C}$. Its temperature is measured every 5 minutes.

t/min	0	5	10	15	20
T/°C	80.0	62.4	50.0	41.2	35.0

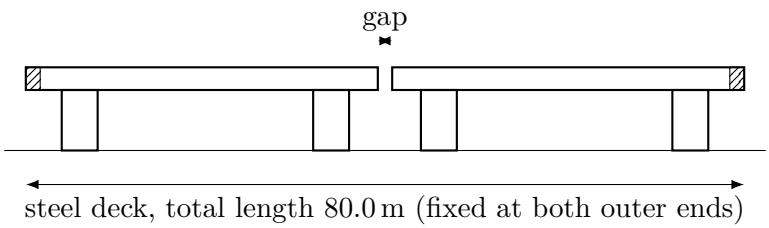
(a) Complete a table of $T - T_{\text{room}}$.

(b) By what factor does $T - T_{\text{room}}$ change in every 10 minutes? What is the half-life of the difference?

(c) Predict T at $t = 30\,\text{min}$.

Level 2 — Standard problems

Exercise 10 — An expansion joint



A steel bridge deck is 80.0 m long in total and is built in two sections with one expansion joint in the middle. The outer ends are fixed, so the two sections expand towards the gap. The joint is built when the temperature is 15 °C. During the year the temperature ranges from −15 °C to 45 °C.

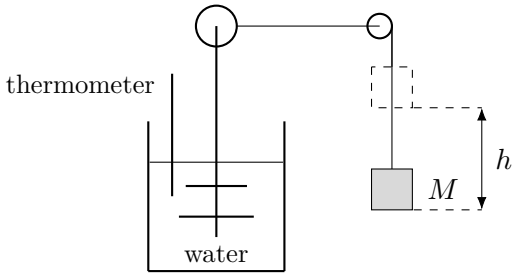
(a) Calculate the total increase in length of the deck from 15 °C to 45 °C.

(b) What is the minimum gap at 15 °C so that the sections never touch?

(c) What is the width of the gap at −15 °C?

(d) The joint is covered by a toothed metal plate. What range of movement, in millimetres, must it allow?

Exercise 11 — Joule's paddle



A mass $M = 2.0\,\text{kg}$ falls through $h = 1.5\,\text{m}$ and, through a pulley, turns a paddle in $m = 0.30\,\text{kg}$ of water in an insulated container. The mass is lifted and let fall 40 times in total. The thermometer reads to 0.1 °C.

(a) Calculate the work done by gravity in the 40 falls.

(b) Calculate the temperature rise of the water if all this work becomes thermal energy of the water.

(c) The measured rise is 0.60 °C. What fraction of the energy reached the water?

(d) Give two reasons for the difference. Why is it better to use 40 falls rather than one?

Exercise 12 — From ice to warm water

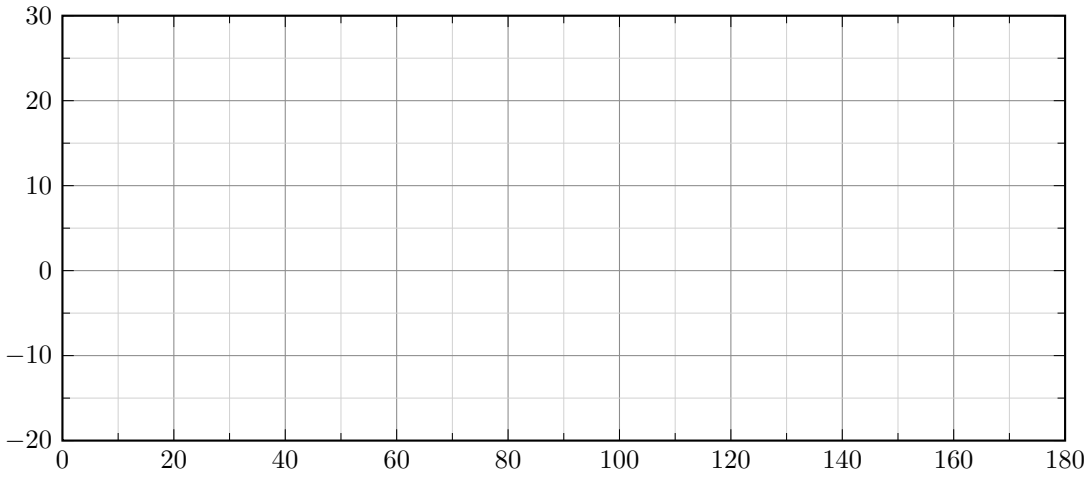
A 500 W heater warms 0.20 kg of ice, initially at −10 °C, until it becomes water at 20 °C. All the energy goes into the ice or water.

(a) Calculate the energy needed for each stage: warming the ice to 0 °C, melting, warming the water to 20 °C.

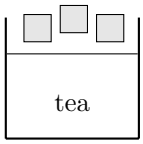
(b) Calculate the time for each stage and the total time.

(c) On the grid, draw the graph of temperature against time.

(d) Explain the shape of the graph at 0 °C.



Exercise 13 — Cooling the tea



0.25 kg at 80 °C
ice cubes, 20 g each, at −10 °C

Hot tea, $m = 0.25$ kg at 80 °C (treat it as water), is cooled by ice cubes from a freezer at −10 °C. Each cube has a mass of 20 g. Ignore the cup.

(a) Calculate the total mass of ice needed to bring the tea to 25 °C, with all the ice melted. How many cubes is this?

(b) Six cubes are added. Is the final temperature above or below 25 °C? Calculate it.

Exercise 14 — Calibrating the Arduino thermometer

A group calibrates a thermistor circuit. The Arduino gives a number N. In melting ice (0 °C) $N = 205$ and in a hot bath at 45 °C (reference thermometer) $N = 555$. They assume that N is a straight-line function of the temperature, $N = N_0 + S\,T$.

(a) Find N_0 and the gradient S.

(b) The sensor now gives $N = 410$. What is the temperature?

(c) Three independent baths are measured, the reference thermometer reading to 0.5 °C:

Bath	1	2	3
Reference temperature /°C	12.0	22.0	33.0
N	296	378	466

Calculate the temperature given by the calibration line for each bath and the difference from the reference.

(d) The client asks for an accuracy of ±1 °C between 0 °C and 45 °C. Do these results prove that the requirement is met? Give one thing you would do to be more confident.

Exercise 15 — Specific heat capacity of a metal block

A student heats an aluminium block of mass 1.00 kg with an electric heater of constant power 50.0 W (measured with a joulemeter) and records its temperature every minute.

t/s	0	60	120	180	240	300	360
T/°C	20.0	22.9	26.1	29.0	35.5	35.2	38.1

(a) State the independent variable, the dependent variable and two variables that must be kept constant.

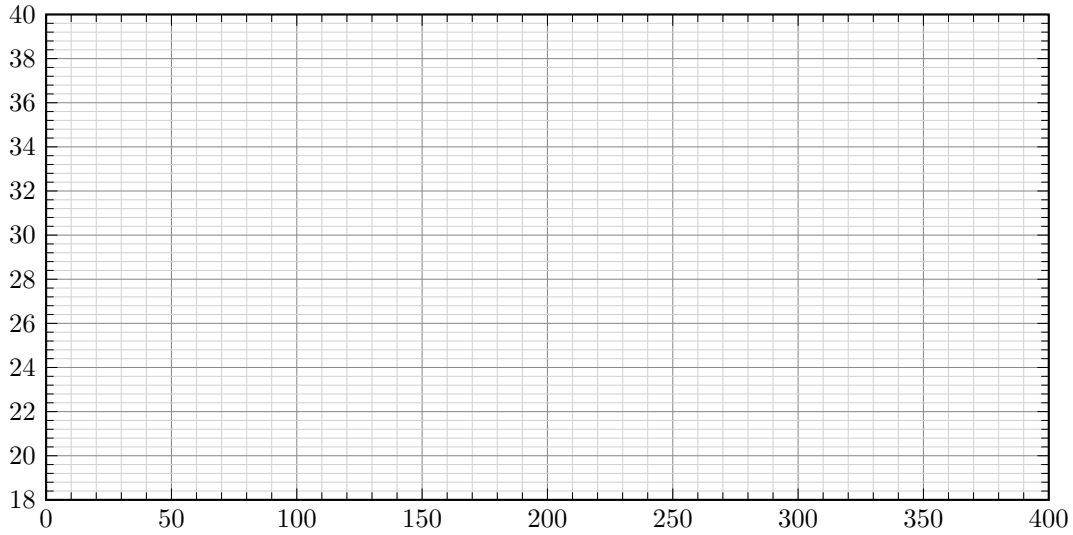
(b) One reading is anomalous. Which one? Give a reason.

(c) Plot the graph of T against t on the grid (axes labelled with quantity and unit) and draw the line of best fit.

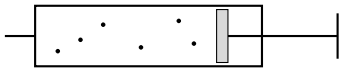
(d) Find the gradient, showing the triangle on the graph.

(e) Use $Pt = mc\,\Delta T$ to find c.

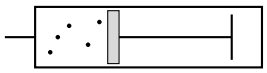
(f) Your result differs from the table value (900 J/(kg °C)). Is your value higher or lower? Explain, and suggest one improvement of the method.



Exercise 16 — A syringe of air



$V_1 = 60\,\text{cm}^3,\ p_1 = 1.0 \times 10^5\,\text{Pa}$



$V_2 = 24\,\text{cm}^3,\ p_2 = ?$

(a) Air in a syringe, $V_1 = 60\,\text{cm}^3$ at $p_1 = 1.0 \times 10^5\,\text{Pa}$, is compressed *slowly* to $V_2 = 24\,\text{cm}^3$. At constant temperature pV is constant. Calculate p_2.

(b) Explain the rise in pressure in terms of the particles.

(c) The same compression is made *very quickly*, with the nozzle closed. The air becomes warmer. Explain why, and say how p_2 compares with your answer to (a).

Exercise 17 — Sweating

During a hot hour of cycling, 0.50 kg of sweat evaporates from the skin.

(a) Calculate the energy removed from the body by the evaporation.

(b) Calculate the average power of this cooling.

(c) At rest a body releases about 100 W. How many times larger is your answer?

(d) Explain why a wet towel on a windy day cools you more than the same towel in still air.

Exercise 18 — Cooling tea: the straight line

Tea cools in a room at $T_{\text{room}} = 20.0\,^\circ\text{C}$.

t/min	0	5	10	15	20
T/°C	85.0	70.6	59.4	50.7	43.9

(a) Complete a table of $T - T_{\text{room}}$ and $\ln(T - T_{\text{room}})$ (3 decimal places).

(b) Is the graph of $\ln(T - T_{\text{room}})$ against t a straight line? Use the values to explain.

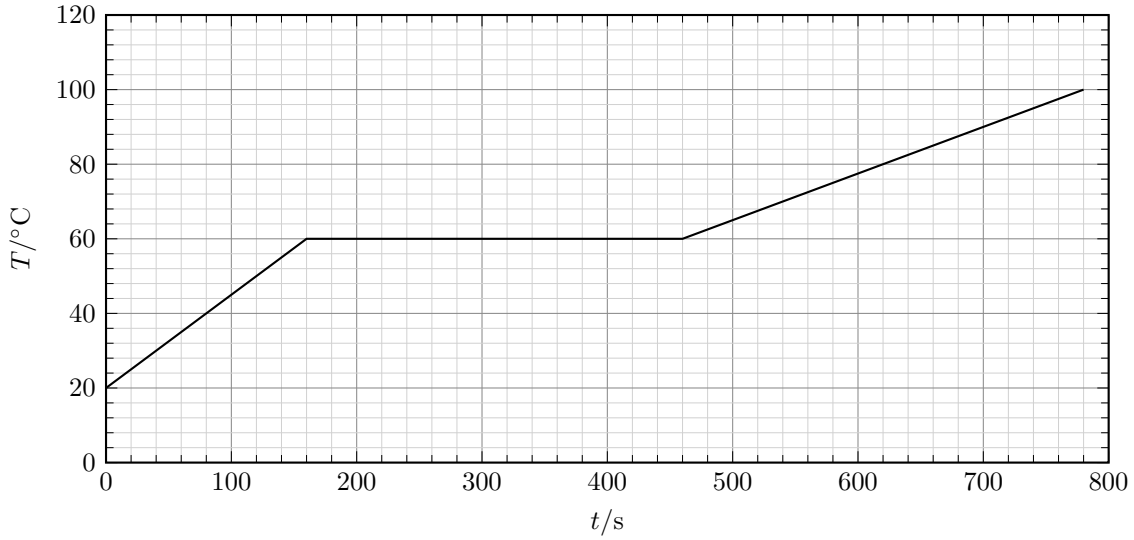
(c) Find the gradient between $t = 0$ and $t = 20\,\text{min}$ and the value of k in 1/min.

(d) Calculate the half-life of the difference $T - T_{\text{room}}$, and the temperature of the tea after three half-lives.

Level 3 — Challenging problems

Exercise 19 — Substance X

A solid sample of a pure substance X, of mass 0.40 kg, is heated by a 200 W heater. The temperature is recorded against time.



(a) Find the melting point of X.

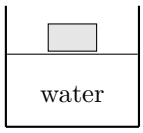
(b) Use the graph to find the specific heat capacity of the solid.

(c) Use the graph to find the specific latent heat of fusion of X.

(d) Find the specific heat capacity of the liquid and explain why the line is less steep than for the solid.

(e) The heater is actually only 90% efficient. Are your values in (b)–(d) too large or too small? Explain.

Exercise 20 — The ice budget



water 0.30 kg at 30 °C
ice 0.15 kg at −10 °C

A block of ice of mass 0.15 kg at −10 °C is dropped into 0.30 kg of water at 30 °C in an insulated container.

(a) How much energy can the water release by cooling to 0 °C?

(b) How much energy does the ice need to warm to 0 °C, and how much to melt completely?

(c) Decide what the final state is, and justify it by comparing the numbers.

(d) Calculate the mass of ice left (or, if all the ice melts, the final temperature).

(e) Repeat the calculation with a block of ice of mass 0.080 kg instead of 0.15 kg.

Exercise 21 — A tape measure in the sun

A steel tape measure is correctly graduated at 20 °C. It is used on a hot day, at 35 °C, to measure a distance, and reads 50.000 m.

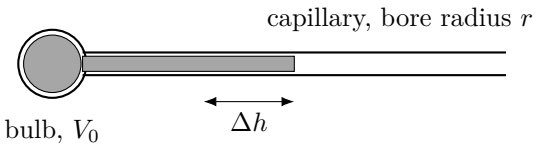
(a) Does the tape read too much or too little? Explain without calculation.

(b) Calculate the true distance.

(c) On a cold day, at −5 °C, the same tape reads 50.000 m. What is the true distance?

(d) The tape is read to 1 mm. Is the error at 35 °C significant? How would the error change for an aluminium tape?

Exercise 22 — Why a thermometer is sensitive



A mercury-in-glass thermometer has a bulb of volume $V_0 = 500\,\text{mm}^3$ and a capillary of radius $r = 0.10\,\text{mm}$. Mercury has $\gamma = 1.8 \times 10^{-4}/^\circ\text{C}$; for glass use $\gamma \approx 3\alpha$ with $\alpha = 9 \times 10^{-6}/^\circ\text{C}$.

(a) Calculate the volume increase of the mercury for a rise of $1\,^\circ\text{C}$, if the glass did not expand. What would Δh be?

(b) The glass bulb also expands. Calculate the *net* volume of mercury that moves into the capillary for $1\,^\circ\text{C}$ and the movement Δh of the column.

(c) By what percentage is the answer of (a) wrong?

(d) A thermometer with a capillary of radius 0.05 mm is made. Compare the sensitivity, and say what limits this design.

Exercise 23 — An electric kettle

A 2.0 kW kettle contains 1.5 kg of water at $20\,^\circ\text{C}$. It reaches the boil ($100\,^\circ\text{C}$) after 290 s.

(a) Calculate the thermal energy gained by the water.

(b) Calculate the electrical energy supplied and the efficiency of the kettle.

(c) The kettle is left on for another 120 s after boiling. Assume that all the electrical energy is now used to boil the water, and calculate the mass of water that turns to steam.

(d) Where does the rest of the energy in (b) go?

Exercise 24 — Cooling tea: can we trust the prediction?

Use the data of Exercise 18 ($T_{\text{room}} = 20.0\,^\circ\text{C}$, best-fit line $\ln(T - T_{\text{room}}) = 4.174 - 0.0500\,t$, t in minutes).

(a) At what time does the tea reach $40\,^\circ\text{C}$? Use the line.

(b) Predict the temperature at $t = 60\,\text{min}$.

(c) A second student takes $T_{\text{room}} = 18.0\,^\circ\text{C}$ by mistake. Using the rows $t = 0$ and $t = 20\,\text{min}$ of the table, calculate their gradient and intercept, and their predicted temperature at $t = 10\,\text{min}$ and at $t = 60\,\text{min}$.

(d) Compare the two students' predictions at 10 min and at 60 min with the correct line. Which prediction is impossible in the long run? Why does the error matter more far from the data?

Exercise 25 — Mixed Paper 2 questions

1. How much energy is needed to raise the temperature of 2.0 kg of a liquid, $c = 2500\,\text{J/(kg}\,^\circ\text{C)}$, from 20 °C to 30 °C?

   A 5000 J
   B 25 000 J
   C 50 000 J
   D 150 000 J

2. A 3.0 kW heater warms 1.0 kg of water by 10 °C. How long does it take?

   A 14 s
   B 42 s
   C 140 s
   D 12 600 s

3. A steel bar 4.0 m long expands by 3.0 mm when heated by ΔT. A steel bar 2.0 m long is heated by $2\,\Delta T$. By how much does it expand?

   A 1.5 mm
   B 3.0 mm
   C 6.0 mm
   D 12 mm

4. A pure solid is heated at constant power. While it is melting its temperature stays constant. Why?

   A The energy supplied makes the particles move faster.
   B The energy supplied is lost to the surroundings.
   C The thermometer cannot follow the change.
   D The energy supplied is used to overcome the forces between the particles, not to increase their speed.

5. 2.0 kg of water at 80 °C is mixed with 1.0 kg of water at 20 °C in an insulated container. What is the final temperature?

   A 50 °C
   B 55 °C
   C 60 °C
   D 70 °C